LLM-assisted specimen; source attributions retain their original authorship. See PROVENANCE.md and the tagged Clean sources.

Clean statement specimens

Prototype 0.1

These are statement renderings, not proof certificates. The collection specializes Mathlib results for syntax comparison. The algorithms section contains CLR-aligned sorting results and number-theoretic supporting lemmas, not a formalization of CLR.

Contents

1 Stewart example **2**

2 Linear algebra **3**

3 Graph theory **6**

4 Analysis **9**

5 Undergraduate algorithms **12**

6 References **15**

1 Stewart example

Result 1 (similarity transforms eigenvector). [Stewart 2001, Theorem 1.10, page 9]
Let n be a natural number. Let A and S be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Let μ be a complex number. Assume $\det(S) \neq 0$ (`invertibility`). Assume $x \neq 0$ (`nonzero`). Assume $A\,x = \mu\,x$ (`eigenpair`). Let $y = S^{-1}\,x$. Then

$$y \neq 0 \text{ and } S^{-1}\,A\,S\,y = \mu\,y.$$

Adapted from the cited source. Scope: eigenvector part.

2 Linear algebra

Result 2 (determinant product). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(A)\,\det(B).$$

Result 3 (determinant transpose). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\det(A^{\mathsf{T}}) = \det(A).$$

Result 4 (determinant product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(B\,A).$$

Result 5 (determinant left swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,(B\,C)) = \det(B\,(A\,C)).$$

Result 6 (determinant right swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B\,C) = \det(A\,C\,B).$$

Result 7 (determinant power). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let k be a natural number. Then

$$\det(A^k) = \det(A)^k.$$

Result 8 (trace sum). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B).$$

Result 9 (trace difference). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A - B) = \operatorname{tr}(A) - \operatorname{tr}(B).$$

Result 10 (trace negative). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}(-A) = -\operatorname{tr}(A).$$

Result 11 (trace transpose). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}(A^{\mathsf{T}}) = \operatorname{tr}(A).$$

Result 12 (trace product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A\,B) = \operatorname{tr}(B\,A).$$

Result 13 (trace cycle). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A\,B\,C) = \operatorname{tr}(C\,A\,B).$$

Result 14 (matrix vector associativity). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$A\,(B\,x) = A\,B\,x.$$

Result 15 (matrix vector addition). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x and y be n-vectors over $\mathbb{C}$. Then

$$A\,(x + y) = A\,x + A\,y.$$

Result 16 (matrix addition action). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$(A + B)\,x = A\,x + B\,x.$$

Result 17 (matrix vector subtraction). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x and y be n-vectors over $\mathbb{C}$. Then

$$A\,(x - y) = A\,x - A\,y.$$

Result 18 (matrix subtraction action). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$(A - B)\,x = A\,x - B\,x.$$

Result 19 (matrix scalar action). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Let c be a complex number. Then

$$A\,(c\,x) = c\,(A\,x).$$

Result 20 (transpose involution). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$A^{\mathsf{T}\mathsf{T}} = A.$$

Result 21 (transpose product). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$A\,B^{\mathsf{T}} = B^{\mathsf{T}}\,A^{\mathsf{T}}.$$

Result 22 (trace commutator). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}([A, B]) = 0.$$

Adapted from the cited source.

Result 23 (trace matrix square). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}((A)^2) = \operatorname{tr}(A\,A).$$

Adapted from the cited source.

Result 24 (determinant matrix square). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\det((A)^2) = \det(A)\ \det(A).$$

Adapted from the cited source.

3 Graph theory

Result 25 (adjacency symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$v \text{ is adjacent to } u \text{ in } G.$$

Result 26 (adjacency commutes). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } G \text{ if and only if } v \text{ is adjacent to } u \text{ in } G.$$

Result 27 (no self edges). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$\text{not } (u \text{ is adjacent to } u \text{ in } G).$$

Result 28 (adjacent vertices distinct). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$u \neq v.$$

Result 29 (union adjacency). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } (G \cup H) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G \text{ or } u \text{ is adjacent to } v \text{ in } H.$$

Result 30 (intersection adjacency). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } (G \cap H) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G \text{ and } u \text{ is adjacent to } v \text{ in } H.$$

Result 31 (complement adjacency). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } \overline{G} \text{ if and only if } u \neq v \text{ and not } (u \text{ is adjacent to } v \text{ in } G).$$

Result 32 (support monotonicity). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Assume G is a subgraph of H (`inclusion`). Then

$$\text{supp}(G) \text{ is a subset of } \text{supp}(H).$$

Result 33 (neighbors in support). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \text{ is a subset of } \operatorname{supp}(G).$$

Result 34 (neighbor membership). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$v \text{ belongs to } N_G(u) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G.$$

Result 35 (not own neighbor). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$\text{not } (u \text{ belongs to } N_G(u)).$$

Result 36 (common neighbors intersection). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) = N_G(u) \cap N_G(v).$$

Result 37 (common neighbors membership). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u, v and w be vertices in $\{0, \ldots, n-1\}$. Then

$$w \text{ belongs to } N_G(u) \cap N_G(v) \text{ if and only if } u \text{ is adjacent to } w \text{ in } G \text{ and } v \text{ is adjacent to } w \text{ in } G.$$

Result 38 (common neighbors symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) = N_G(v) \cap N_G(u).$$

Result 39 (common neighbors left). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) \text{ is a subset of } N_G(u).$$

Result 40 (common neighbors right). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) \text{ is a subset of } N_G(v).$$

Result 41 (reachability reflexive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$u \text{ is reachable from } u \text{ in } G.$$

Result 42 (reachability symmetric). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume v is reachable from u in G (`path`). Then

$$u \text{ is reachable from } v \text{ in } G.$$

Result 43 (reachability transitive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u, v and w be vertices in $\{0, \ldots, n-1\}$. Assume v is reachable from u in G (`first`). Assume w is reachable from v in G (`second`). Then

$$w \text{ is reachable from } u \text{ in } G.$$

Result 44 (edge implies reachability). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$v \text{ is reachable from } u \text{ in } G.$$

Result 45 (nonadjacency symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ and } v \text{ are nonadjacent in } G \text{ if and only if } v \text{ and } u \text{ are nonadjacent in } G.$$

Adapted from the cited source.

Result 46 (nonadjacency reflexive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$u \text{ and } u \text{ are nonadjacent in } G.$$

Adapted from the cited source.

Result 47 (complement from nonadjacency). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } \overline{G} \text{ if and only if } u \neq v \text{ and } u \text{ and } v \text{ are nonadjacent in } G.$$

Adapted from the cited source.

4 Analysis

Result 48 (exponential positive). [Mathlib]
Let x be a real number. Then

$$0 < \exp(x).$$

Result 49 (exponential nonzero). [Mathlib]
Let x be a real number. Then

$$\exp(x) \neq 0.$$

Result 50 (exponential sum). [Mathlib]
Let x and y be real numbers. Then

$$\exp(x + y) = \exp(x)\ \exp(y).$$

Result 51 (exponential difference). [Mathlib]
Let x and y be real numbers. Then

$$\exp(x - y) = \frac{\exp(x)}{\exp(y)}.$$

Result 52 (logarithm exponential). [Mathlib]
Let x be a real number. Then

$$\log(\exp(x)) = x.$$

Result 53 (exponential logarithm). [Mathlib]
Let x be a real number. Assume $0 < x$ (`positive`). Then

$$\exp(\log(x)) = x.$$

Result 54 (logarithm product). [Mathlib]
Let x and y be real numbers. Assume $x \neq 0$ (`nonzero_x`). Assume $y \neq 0$ (`nonzero_y`). Then

$$\log(x\,y) = \log(x) + \log(y).$$

Result 55 (logarithm quotient). [Mathlib]
Let x and y be real numbers. Assume $x \neq 0$ (`nonzero_x`). Assume $y \neq 0$ (`nonzero_y`). Then

$$\log(\frac{x}{y}) = \log(x) - \log(y).$$

Result 56 (square root nonnegative). [Mathlib]
Let x be a real number. Then

$$0 \leq \sqrt{x}.$$

Result 57 (square of square root). [Mathlib]
Let x be a real number. Assume $0 \leq x$ (`nonnegative`). Then

$$\sqrt{x}^2 = x.$$

Result 58 (square root of square). [Mathlib]
Let x be a real number. Then

$$\sqrt{x^2} = |x|.$$

Result 59 (square root product). [Mathlib]
Let x and y be real numbers. Assume $0 \leq x$ (`nonnegative`). Then

$$\sqrt{x\,y} = \sqrt{x}\,\sqrt{y}.$$

Result 60 (trigonometric identity). [Mathlib]
Let x be a real number. Then

$$\sin(x)^2 + \cos(x)^2 = 1.$$

Result 61 (sine addition). [Mathlib]
Let x and y be real numbers. Then

$$\sin(x + y) = \sin(x)\,\cos(y) + \cos(x)\,\sin(y).$$

Result 62 (cosine addition). [Mathlib]
Let x and y be real numbers. Then

$$\cos(x + y) = \cos(x)\,\cos(y) - \sin(x)\,\sin(y).$$

Result 63 (sine odd). [Mathlib]
Let x be a real number. Then

$$\sin(-x) = -\sin(x).$$

Result 64 (cosine even). [Mathlib]
Let x be a real number. Then

$$\cos(-x) = \cos(x).$$

Result 65 (continuous composition). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$(f \circ g) \text{ is continuous.}$$

Result 66 (continuous sum). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$f + g \text{ is continuous.}$$

Result 67 (continuous product). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$f\ g \text{ is continuous.}$$

Result 68 (authored square root). [Mathlib]
Let x be a real number. Then

$$\sqrt{(x)^2} = |x|.$$

Adapted from the cited source.

Result 69 (authored trigonometric energy). [Mathlib]
Let x be a real number. Then

$$\text{trigonometric energy}(x) = 1.$$

Adapted from the cited source.

Result 70 (distance square root). [Mathlib]
Let x and y be real numbers. Then

$$\sqrt{\text{squared distance}(x, y)} = |x - y|.$$

Adapted from the cited source.

5 Undergraduate algorithms

Result 71 (insertion sort sorted). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{insertionSort}(xs) \text{ is sorted.}$$

Result 72 (insertion sort permutation). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{insertionSort}(xs) \text{ is a permutation of } xs.$$

Result 73 (insertion sort length). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$|\text{insertionSort}(xs)| = |xs|.$$

Result 74 (insertion sort membership). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$a \text{ belongs to } \text{insertionSort}(xs) \text{ if and only if } a \text{ belongs to } xs.$$

Result 75 (sorted input unchanged). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Assume xs is sorted (`ordered`). Then

$$\text{insertionSort}(xs) = xs.$$

Result 76 (ordered insert length). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$|\text{insert}(a, xs)| = |xs| + 1.$$

Result 77 (ordered insert membership). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a and b be natural numbers. Then

$$b \text{ belongs to } \text{insert}(a, xs) \text{ if and only if } b = a \text{ or } b \text{ belongs to } xs.$$

Result 78 (ordered insert permutation). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$\text{insert}(a, xs) \text{ is a permutation of } (a :: xs).$$

Result 79 (ordered insert preserves sortedness). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Assume xs is sorted (`ordered`). Then

$$\text{insert}(a, xs) \text{ is sorted.}$$

Result 80 (merge sort sorted). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{mergeSort}(xs) \text{ is sorted.}$$

Result 81 (merge sort sorted input). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Assume xs is sorted (`ordered`). Then

$$\text{mergeSort}(xs) = xs.$$

Result 82 (sorting algorithms agree). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{mergeSort}(xs) = \text{insertionSort}(xs).$$

Result 83 (merge preserves sortedness). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume xs is sorted (`hx`). Assume ys is sorted (`hy`). Then

$$\text{merge}(xs, ys) \text{ is sorted.}$$

Result 84 (insertion sort subsequence). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume ys is sorted (`ordered`). Assume ys is a sublist of xs (`subsequence`). Then

$$ys \text{ is a sublist of } \text{insertionSort}(xs).$$

Result 85 (ordered insert contains input). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$xs \text{ is a sublist of } \text{insert}(a, xs).$$

Result 86 (euclidean coprime reduction). [Mathlib]
Let a, b and k be natural numbers. Then

$$a \text{ and } b + k\,a \text{ are coprime if and only if } a \text{ and } b \text{ are coprime.}$$

Result 87 (euclidean coprime subtraction). [Mathlib]
Let a and b be natural numbers. Assume $a \leq b$ (`ordered`). Then

$$b - a \text{ and } a \text{ are coprime if and only if } b \text{ and } a \text{ are coprime.}$$

Result 88 (coprime lcm product). [Mathlib]
Let a and b be natural numbers. Assume a and b are coprime (`cop`). Then

$$\text{lcm}(a, b) = a\,b.$$

Result 89 (common divisor of coprimes). [Mathlib]
Let a, b and k be natural numbers. Assume a and b are coprime (`cop`). Assume k divides a (`divides_a`). Assume k divides b (`divides_b`). Then

$$k = 1.$$

Result 90 (coprime product divisibility). [Mathlib]
Let a, b and k be natural numbers. Assume k and b are coprime (`cop`). Then

$$k \text{ divides } a\,b \text{ if and only if } k \text{ divides } a.$$

Result 91 (insertion sort certificate). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{sorting certificate}(xs, \text{insertionSort}(xs)).$$

Adapted from the cited source.

Result 92 (merge sort certificate). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{sorting certificate}(xs, \text{mergeSort}(xs)).$$

Adapted from the cited source.

Result 93 (certified sort unchanged). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume sorting certificate(xs, ys) (`certificate`). Then

$$\text{insertionSort}(ys) = ys.$$

Adapted from the cited source.

6 References

Mathlib The Mathlib Community. The Lean Mathematical Library. Version 4.24.0. Revision f897ebcf72cd. Source

Stewart 2001 G. W. Stewart. Matrix Algorithms, Volume II: Eigensystems. 2001. SIAM.